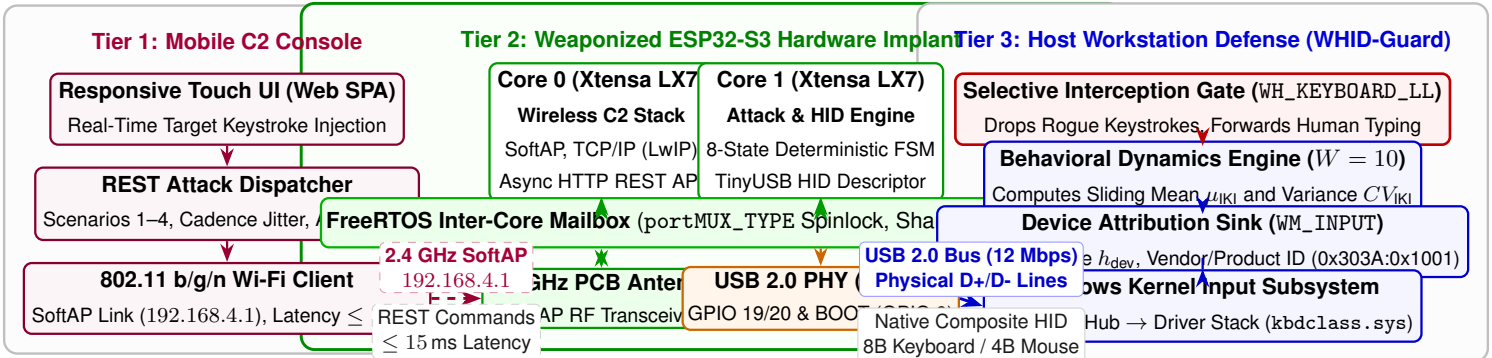

Tier 1: Mobile C2 Console
Responsive Touch UI (Web SPA)
Real-Time Target Keystroke Injection
REST Attack Dispatcher
Scenarios 1–4, Cadence Jitter,
802.11 b/g/n Wi-Fi Client
SoftAP Link (192.168.4.1), Latency ≤
Tier 2: Weaponized ESP32-S3 Hardware Implant
Core 0 (Xtensa LX7
Wireless C2 Stack
SoftAP, TCP/IP (LwIP)
Async HTTP REST AP
Core 1 (Xtensa LX7)
Attack & HID Engine
8-State Deterministic FSM
TinyUSB HID Descriptor
FreeRTOS Inter-Core Mailbox (portMUX_TYPE Spinlock, Sha
2.4 GHz SoftAP
192.168.4.1
REST Commands
≤ 15 ms Latency
GHz PCB Anter
AP RF Transceiv
USB 2.0 PHY (
GPIO 19/20 & BOO
USB 2.0 Bus (12 Mbps)
Physical D+/D- Lines
Native Composite HID
8B Keyboard / 4B Mouse
Tier 3: Host Workstation Defense (WHID-Guard)
Selective Interception Gate (WH_KEYBOARD_LL)
Drops Rogue Keystrokes, Forwards Human Typing
Behavioral Dynamics Engine ($W = 10$)
Computes Sliding Mean μ_{IKI} and Variance CV_{IKI}
Device Attribution Sink (WM_INPUT)
e h_{dev}, Vendor/Product ID (0x303A:0x1001)
ows Kernel Input Subsystem
Hub → Driver Stack (kbdclass.sys)